

Ultra-High-Resolution Silicon Photonic Beamformer via Path-Reversal Optical True-Time Delay

Senyu Zhang⁽¹⁾⁽²⁾, Yifei Chong⁽¹⁾⁽²⁾, Wei Luo⁽¹⁾⁽²⁾, Yunlong Li⁽¹⁾⁽²⁾, Shuang Zheng^{(1)(2)(3)*}, Minming Zhang^{(1)(2)(3)*}

⁽¹⁾ School of Optical and Electronic Information and Wuhan National Laboratory for Optoelectronics, Huazhong University of Science and Technology, Wuhan, Hubei 430074, China

⁽²⁾ National Engineering Research Center for Next Generation Internet Access System, Wuhan, Hubei 430074, China

⁽³⁾ Optics Valley Laboratory, Wuhan, Hubei 430074, China

*zshust@hust.edu.cn and *mmz@hust.edu.cn

Abstract We demonstrate a silicon photonic beamformer employing a path-reversal optical true-time delay mechanism, increasing squint-free pointing angles in switched M-bit architectures from 2^M to $2^{M+1}-1$ without additional delay stages. Our 5-bit chip achieves 63 angles over 35 GHz with <0.5 ps delay error. ©2026 The Author(s)

Introduction

Microwave phased-array antennas (PAAs) are essential for high-capacity wireless communications and advanced radar systems. Compared with resonance-based architectures such as microring resonators (MRRs), which suffer from limited operating bandwidth and require complex thermal stabilization, switched optical true-time delay lines (OTTDLs) provide a robust alternative with broader bandwidth and lower sensitivity to temperature fluctuations. However, their discrete tuning nature inherently limits the number of accessible pointing angles and creates a trade-off between angular resolution and broadband operation [1-3].

A parallel M-bit OTTDL architecture reported previously enabled broadband beamforming, but it intrinsically suffered from beam squint for negative pointing angles because negative incremental delays were unavailable [2]. More recently, pre-compensation schemes were introduced to support squint-free steering for both positive and negative angles [4]. Nevertheless, in such

switched M-bit architectures, the number of squint-free pointing angles remains limited to 2^M . Further improving the angular resolution therefore generally requires additional physical delay stages, which inevitably increase insertion loss, chip footprint, and control complexity [5-9].

In this paper, we propose and experimentally characterize an ultra-high-resolution silicon photonic beamformer based on path-reversal optical true-time delay. By introducing a path-reversal routing mechanism, the proposed architecture increases the number of squint-free pointing angles in switched M-bit architectures from 2^M to $2^{M+1}-1$ without introducing extra delay stages. The fabricated 5-bit prototype supports 63 pointing angles over 35 GHz with a delay error below 0.5 ps.

Operation Principle

As shown in Fig. 1, the proposed 1×8 beamformer employs $N=8$ parallel OTTDLs. In this high-resolution architecture, the delay increment is intentionally designed to increase linearly from

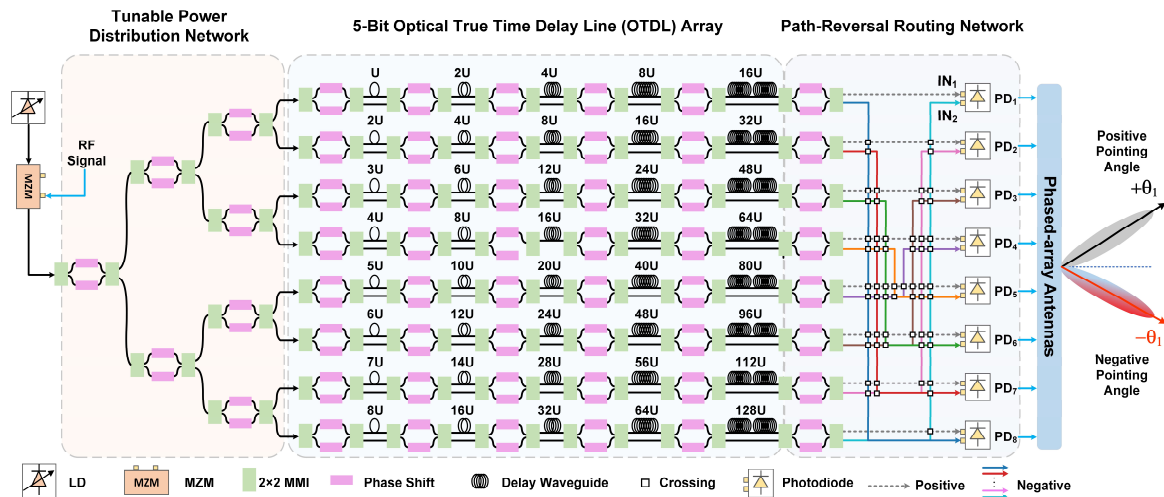


Fig. 1: Schematic of the proposed 1×8 high-resolution silicon photonic beamformer, consisting of parallel 5-bit OTTDLs with linearly increasing delay increments and a path-reversal routing network.

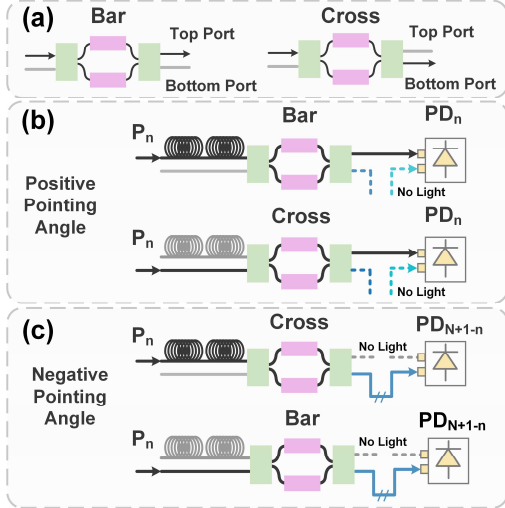


Fig. 2: Operating principle of the path-reversal mechanism for increasing the number of pointing angles. (a) Two routing states of the final MZI switch. (b) Positive pointing operation. (c) Negative pointing operation enabled by path reversal.

the first channel to the N -th channel, rather than remaining constant across the array. For a given M -bit control word $B \in [0, 2^M - 1]$, the relative delay of the n -th OTTDL is expressed as $T_n = nBU$, where $U = 0.52$ ps is the unit delay.

The key element enabling the resolution enhancement is the path-reversal network placed before the dual-port photodetector (PD) array, as illustrated in Fig. 2. This network consists of a column of 2×2 Mach-Zehnder interferometer (MZI) switches together with an integrated waveguide-crossing manifold, and supports two routing configurations for positive and negative pointing operation.

For positive pointing angles, the optical signals are routed from the n -th delay path to the top

port and are directly detected by the corresponding n -th dual-port photodetector, PD_n . In this case, the programmed positive delay gradient across the array is preserved.

For negative pointing angles, the optical signals are routed to the bottom ports and then redirected by the integrated waveguide-crossing network from the n -th optical path to the $(N+1-n)$ -th photodetector, PD_{N+1-n} . This physical port inversion reverses the delay ordering across the antenna channels and thereby produces an effective negative delay gradient without requiring negative physical delay lines.

By mapping the $2^M - 1$ nonzero delay states to both positive and negative pointing directions, and including the broadside state ($B=0$), the total number of accessible squint-free pointing angles becomes $2^{M+1} - 1$. For the 5-bit prototype ($M=5$), the proposed architecture therefore provides 63 squint-free pointing angles without increasing the number of physical delay stages.

Device Fabrication and Characterization

The beamformer was fabricated on a standard SOI platform. Figures 3(a) and 3(b) show the chip micrograph and the packaged device, respectively. The integrated 1×8 tunable power splitter enables independent adjustment of the optical power in the eight channels. This capability not only supports arbitrary amplitude tapering schemes, such as Chebyshev or Taylor windowing for sidelobe suppression, but also helps compensate for slight insertion-loss imbalances introduced by the crossing network during negative-angle operation.

The constituent devices exhibit favorable measured performance. The 2×2 MZI switches,

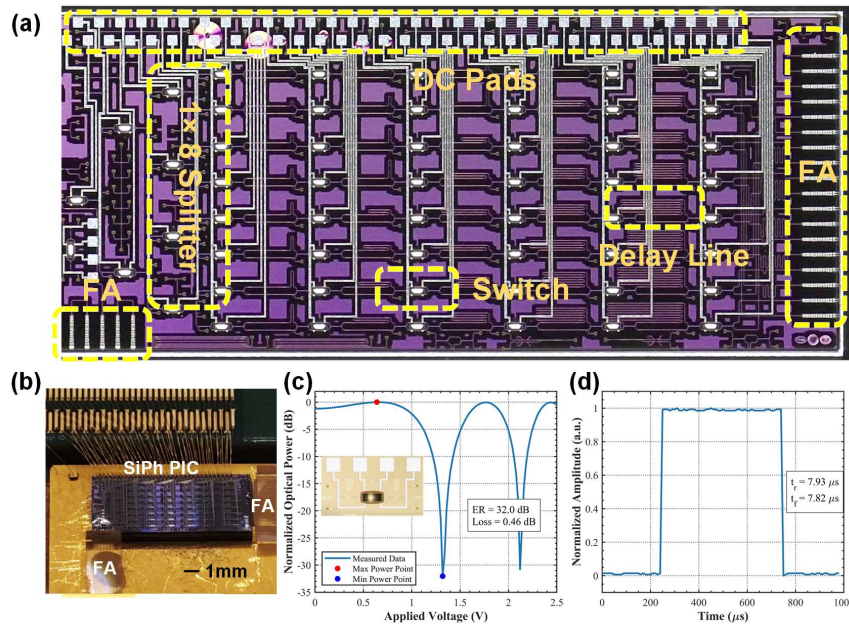


Fig. 3: Device characterization of the fabricated silicon photonic beamformer. (a) Chip micrograph. (b) Packaged device. (c) Measured MZI-switch response showing a 32.0 dB extinction ratio and a 0.46 dB insertion loss. (d) Measured switching response with a 7.9 μs rise/fall time.

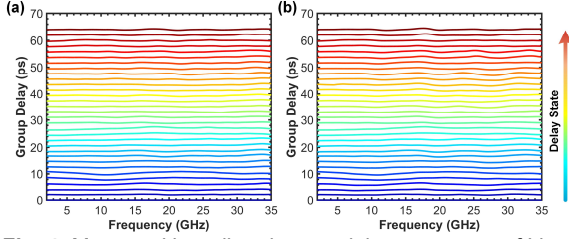


Fig. 4: Measured broadband group-delay responses of Line 5 from 2 to 35 GHz. (a) Positive delay states routed through the top-port path. (b) Negative delay states routed through the bottom-port path and the integrated crossing manifold.

implemented using optimized multimode interferometers (MMIs), achieve an extinction ratio of 32.0 dB and an insertion loss of 0.46 dB. The measured thermo-optic switching time is approximately 7.9 μ s, enabling rapid beam reconfiguration. The complete 8-channel, 5-bit beamformer is compactly integrated on a single die with an active footprint of only 3×8 mm².

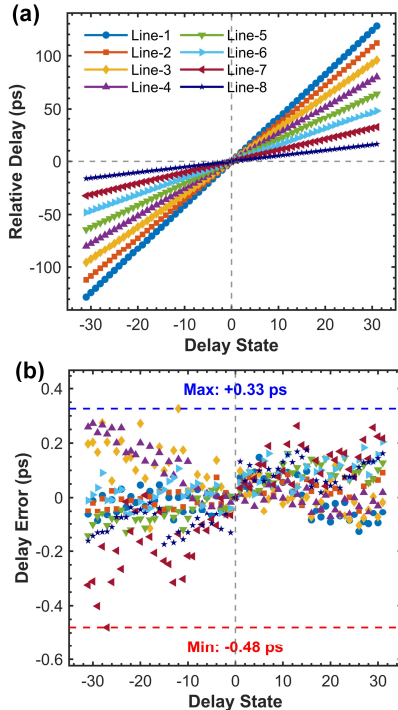


Fig. 5: Measured delay performance of the 8-channel beamformer. (a) Average relative delay as a function of configuration for all eight channels. (b) Delay-error distribution.

Experimental Results

To validate the broadband delay performance and the functionality of the path-reversal mechanism, the chip was characterized using a vector network analyzer (VNA) over the frequency range from 2 to 35 GHz. Figure 4 presents the measured group-delay responses of a representative channel (Line 5). Figure 4(a) shows 15 positive delay states routed through the top-port path, whereas Fig. 4(b) shows 15 negative delay states routed through the bottom-port path and the integrated crossing manifold. Both routing directions

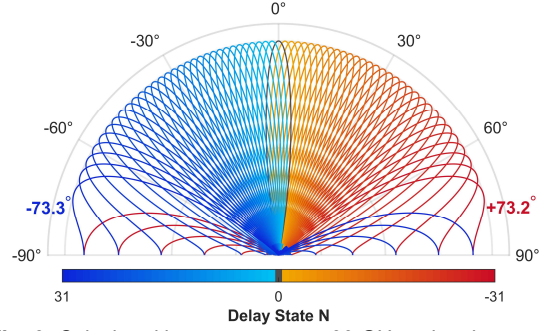


Fig. 6: Calculated beam patterns at 30 GHz using the experimentally measured true-time delays for an antenna spacing of $d = 4.3$ mm. The measured delays support 63 squint-free pointing angles from -73.3° to $+73.2^\circ$.

exhibit similarly flat delay responses over the full measurement band, indicating that the path-reversal network introduces no observable degradation in the broadband delay characteristics.

The time delays of all eight channels were extracted by averaging the measured group delay over the operating frequency range. As shown in Fig. 5(a), the relative delays maintain excellent linearity across all 63 configurations. The corresponding delay-error distribution in Fig. 5(b) shows that the delay deviation for all channels and configurations is bounded between -0.48 ps and $+0.33$ ps.

The beam-pointing capability was further evaluated using a semi-empirical model. Specifically, the experimentally measured time delays were applied to an ideal linear antenna array with an element spacing of $d=4.3$ mm, and the array factor was calculated at 30 GHz. As shown in Fig. 6, the measured true-time delays support 63 dense squint-free pointing angles spanning a wide field of view from -73.3° to $+73.2^\circ$, symmetrically distributed over the negative and positive domains. These results verify that the proposed path-reversal architecture effectively doubles the number of accessible pointing angles in a conventional 5-bit switched beamformer without introducing additional delay stages.

Conclusion

In conclusion, we have proposed and experimentally characterized an ultra-high-resolution silicon photonic beamformer based on path-reversal optical true-time delay. By reversing the delay ordering through an integrated crossing network, the proposed architecture increases the number of squint-free pointing angles in switched M-bit architectures from 2^M to $2^{M+1}-1$ without introducing additional physical delay stages. The fabricated 5-bit prototype supports 63 pointing angles over 35 GHz and exhibits a delay error below 0.5 ps together with a microsecond-level switching speed. These results highlight a compact and scalable route toward broadband photonic beamforming for future radar and wireless communication systems.

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References

- [1] P. Zheng, C. Wang, X. Xu, J. Li, D. Lin, G. Hu, R. Zhang, B. Yun, and Y. Cui, "A seven bit silicon optical true time delay line for Ka-band phased array antenna," *IEEE Photonics Journal*, vol. 11, no. 4, pp. 1-9, 2019, DOI: 10.1109/JPHOT.2019.2927487
- [2] C. Zhu, L. Lu, W. Shan, W. Xu, G. Zhou, L. Zhou, and J. Chen, "Silicon integrated microwave photonic beamformer," *Optica*, vol. 7, no. 9, pp. 1162-1170, 2020, DOI: 10.1364/OPTICA.391521
- [3] H. Sun, L. Lu, Y. Liu, Z. Ni, J. Chen, and L. Zhou, "Broadband 1×8 optical beamforming network based on anti-resonant microring delay lines," *Journal of Light-wave Technology*, vol. 40, no. 20, pp. 6919-6928, 2022, DOI: 10.1109/JLT.2022.3175768
- [4] P. Martinez-Carrasco, T. H. Ho, D. Wessel, and J. Capmany, "Ultrabroadband high-resolution silicon RF-photonics beamformer," *Nature Communications*, vol. 15, no. 1433, 2024, DOI: 10.1038/s41467-024-45743-9
- [5] D. Pérez-López, A. López, P. DasMahapatra, and J. Capmany, "Multipurpose self-configuration of programmable photonic circuits," *Nature Communications*, vol. 11, no. 1, pp. 6359, 2020, DOI: 10.1038/s41467-020-19608-w
- [6] Y. Xie, S. Hong, H. Yan, C. Zhang, L. Zhang, L. Zhuang, and D. Dai, "Low-loss chip-scale programmable silicon photonic processor," *Opto-Electronic Advances*, vol. 6, no. 220030, 2023, DOI: 10.29026/oea.2023.220030
- [7] Z. Ni, L. Lu, Y. Liu, Y. Wang, J. Chen, and L. Zhou, "Silicon-Integrated 8-Channel 6-bit Tunable Optical True-Time Delay Lines with High Switching Speed and Low Loss," in *2023 International Topical Meeting on Microwave Photonics (MWP)*, Nanjing, China, 2023, pp. 1-5, DOI: 10.1109/MWP58203.2023.10416568
- [8] S. Hong, J. Wu, Y. Xie, X. Ke, H. Li, L. Lyv, Y. Peng, Q. Yao, Y. Shi, K. Wang, L. Zhuang, P. Wang, and D. Dai, "Versatile parallel signal processing with a scalable silicon photonic chip," *Nature Communications*, vol. 16, no. 288, 2025, DOI: 10.1038/s41467-024-55162-5
- [9] S. Shi, H. Niu, W. Shi, D. Lin, S. Li, S. Pan, and B. Yun, "Integrated optical tunable delay line and microwave photonic beamforming chip: A review," *Laser & Photonics Reviews*, vol. 19, no. 2400663, 2025, DOI: 10.1002/lpor.202400663